

SONOMETER (05)

Aim

To determine the linear density of the sonometer wire and find the unknown mass by resonance method.

Materials and Apparatus

Sonometer, tuning forks, two wooden pegs, paper riders, rubber pad, weights, unknown mass and meter ruler.

Theory

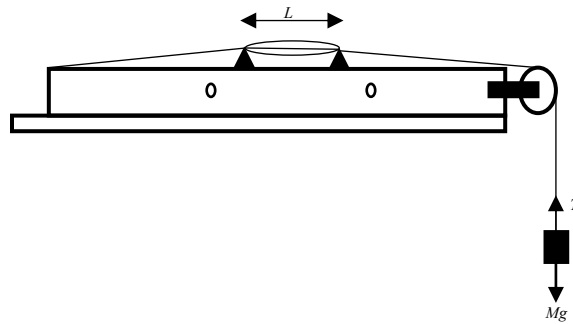


Fig. 1: The schematic diagram of the sonometer experimental setup and the wave pattern at fundamental resonance.

Where, T : tension of the string, M : attached mass, L : resonance length

$$\text{For the fundamental note,} \quad \lambda = 2L \quad (1)$$

$$\text{Velocity of a transverse wave in a stretched string,} \quad V = \sqrt{T/\mu} \quad (2)$$

$$\text{Also,} \quad V = f\lambda \quad (3)$$

$$\text{From equations (1), (2) and (3)} \quad V = \sqrt{Mg/\mu}$$

$$L^2 = \frac{g}{4\mu f^2} M$$

Where,

f : frequency

λ : wavelength

μ : linear density of wire

T : tension of the wire

Procedure

a) Hang an appropriate mass to the sonometer wire.

Q: How do you decide the hanging mass to start with?

b) Measure the resonance length for the first resonance condition, by using given tuning forks.

Q: How do you identify whether the resonance is occurred?

Q: Measure the first resonance length, to determine which tuning fork should be used to continue the experiment with varying attached masses.

c) Use a graphical method to determine the linear density of the sonometer wire.

d) Find the unknown mass.

Q: Suggest an experimental way to find the density of given liquid by using sonometer setup.